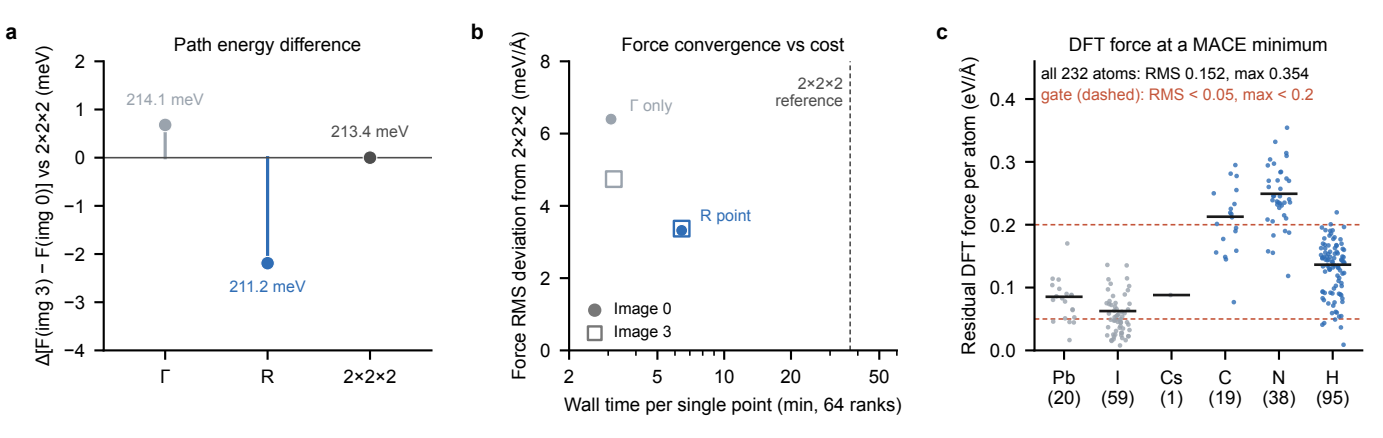
a
Path energy difference
214.1 meV
213.4 meV
211.2 meV
Δ[F(img 3) − F(img 0)] vs 2×2×2 (meV)
2
1
0
−1
−2
−3
−4
Γ
R
2×2×2
b
Force convergence vs cost
Γ only
2×2×2
reference
R point
Image 0
Image 3
Force RMS deviation from 2×2×2 (meV/Å)
8
6
4
2
0
2
5
10
20
50
Wall time per single point (min, 64 ranks)
c
DFT force at a MACE minimum
all 232 atoms: RMS 0.152, max 0.354
gate (dashed): RMS < 0.05, max < 0.2
Residual DFT force per atom (eV/Å)
0.4
0.3
0.2
0.1
0.0
Pb
(20)
I
(59)
Cs
(1)
C
(19)
N
(38)
H
(95)